

Ship Auto Way Maps Points Tracking

Dokumentasi sistem **Ship Auto Way Maps Points Tracking** — kontrol kapal model ESP32-S3 dengan waypoint, tuning algoritma auto, telemetry real-time, dan dashboard PySide6.

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1. Ringkasan sistem

Sistem ini memungkinkan:

- **Monitoring live** posisi, heading, rudder, RPM, baterai dari kapal model
- **Perencanaan waypoint** di peta (Home + hingga 10 waypoint navigasi)
- **Pengiriman waypoint & tuning** ke kapal via ESP-NOW (disimpan di NVS Remote)
- **Verifikasi read-back** parameter tuning (\$TUNGET / \$STACK)
- **Auto track Alg 1** — navigasi waypoint dengan kontrol PD heading
- **Auto track Alg 2** — stub (default, rudder netral)
- **Analisis log CSV** — replay data dengan peta dan plot

Alur end-to-end:

```
Dashboard (PySide6) ■ USB serial 115200, ASCII + CSV ▼ User-Side ESP32-S3
(ESP-Now_ESP32-S3_User-Side-01) ■ ESP-NOW peer-to-peer ▼ Remote-Side ESP32-S3
(ESP-Now_ESP32-S3_Remote-Side-01) — di kapal
```

2. Komponen & path proyek

Komponen	Path (dari root repo)	File utama
Dashboard	Pythonfile/Way_Points_Tracking/	Local Monitor Dashboard-beta1.1.1.py
User-Side	PlatformIO/Way_Points_Tracking/ESP-Now/src/main.cpp	ser-Side-01/
Remote-Side	PlatformIO/Way_Points_Tracking/ESP-Now/src/main.cpp	remote-Side-01/

Dokumentasi ini disalin ke setiap folder proyek firmware sebagai referensi yang sama.

Salinan: ESP-Now_ESP32-S3_Remote-Side-01/, ESP-Now_ESP32-S3_User-Side-01/

3. Arsitektur & alur data

3.1 Telemetry (kapal → PC)

1. Remote membaca sensor ~10 Hz, membentuk struct `DataToSend`
2. Remote mengirim struct via ESP-NOW ke User-Side
3. User-Side mencetak CSV 23-kolom ke USB serial
4. Dashboard `poll_serial()` mem-parse, menampilkan nilai fisik, log CSV

3.2 Waypoint & tuning (PC → kapal)

1. Dashboard validasi Home + waypoint, kirim `$WPSET`
2. User-Side parse, forward ESP-NOW `0xA1` ke Remote
3. Remote simpan waypoint di RAM, ACK `0xC1` → User-Side → `$WACK,OK,WP`
4. Dashboard kirim `$TUNSET` → Remote simpan NVS → `$WACK,OK,TUN`
5. Dashboard kirim `$TUNGET` → Remote balas `0xA3` → `$TACK,...` → verifikasi

3.3 Timeout

Lapisan	Timeout	Catatan
User-Side menunggu Remote ACK/TACK	2.5 s	ACK_TIMEOUT_MS
Dashboard menunggu respons serial	3 s	per langkah WP / TUN / TUNGET

4. Remote-Side (firmware kapal)

Proyek: ESP-Now_ESP32-S3_Remote-Side-01

Firmware di kapal: sensor, actuator, kontrol rudder/propeller, waypoint tracking, NVS tuning, telemetry ESP-NOW.

4.1 Hardware

Perangkat	Interface	Pin / catatan
MCU	ESP32-S3 DevKitC1-N16R8	16 MB Flash, 8 MB PSRAM
RC FS-IA6B	PPM	GPIO 4
Rudder servo	PWM 50 Hz	GPIO 5; feedback ADC GPIO 8, 3
Propeller speed/dir	PWM (Servo lib)	GPIO 6, 7
Encoder prop 1/2	Interrupt	GPIO 9, 10
GNSS u-blox	Serial1	GPIO 17 RX, 18 TX; 115200, 10 Hz
IMU HWT905TTL / JY901	Serial2	GPIO 15 RX, 16 TX; 57600
Baterai 1	ADC	GPIO 1 (divider ×5)
Baterai 2	ADC	GPIO 2 (divider ×5)

4.2 Mapping channel RC

Channel	Fungsi
CH1	Rudder manual (map -40° ... $+40^{\circ}$)
CH3	Kecepatan propeller
CH5	Arah propeller
CH6	Auto/Manual — ≥ 1750 = Auto, < 1750 = Manual

4.3 Mode operasi

Mode	RC CH6	Algoritma NVS	mode_auto	Rudder
Manual	< 1750	—	0	CH1 langsung
Auto Alg 1	≥ 1750	alg = 1	1	PD waypoint tracking
Auto Alg 2	≥ 1750	alg = 2 (default)	2	Netral (stub)

4.4 Penerimaan ESP-NOW

msg_type	Handler	Aksi
0xA1	handleWaypointsPayload	Simpan waypoint RAM, reset g_active_wp_index, ACK WP
0xA2	handleTrackConfigSet	Validasi, apply runtime, simpan NVS, ACK TUN
0xB1	sendTrackConfigResponse	Balas config aktif via 0xA3

4.5 Struct waypoint (waypoints_payload, ~180 byte)

```
uint8_t msg_type = 0xA1 uint8_t home_valid = 1 jika Home valid uint8_t wp_count = 0..10 (waypoint navigasi, tidak termasuk Home) uint8_t reserved double home_lat, home_lon double wp_lat[10], wp_lon[10]
```

Home disimpan terpisah; wp_count hanya menghitung waypoint klik di peta.

4.6 Dependencies PlatformIO

ESP32Servo, TinyGPSPlus, JY901 (lib lokal), Preferences, ESP-NOW, WiFi.

5. User-Side (gateway USB)

Proyek: ESP-Now_ESP32-S3_User-Side-01

Bridge: perintah ASCII dari PC ↔ binary ESP-NOW ke Remote. **Tidak** mengirim \$WACK, OK sebelum Remote membalas 0xC1.

5.1 Peran

- Parse baris serial (\$WPSET, \$TUNSET, \$TUNGET)
- Forward payload ESP-NOW ke Remote
- Tunggu ACK (0xC1) atau tuning response (0xA3)
- Relay telemetry struct Remote sebagai CSV ke USB

5.2 Loop utama

- Baca serial baris-per-baris (buffer, max line length)
- checkAckTimeout() — emit \$WACK, ERR, ... TIMEOUT atau \$TACK, ERR, TIMEOUT
- Saat telemetry diterima: printTelemetryCsv() langsung

5.3 Struct telemetry (receivedfromremoteside)

Identik field dengan DatatoSend Remote (64 byte). User-Side tidak mengubah skala — dashboard yang decode.

6. Dashboard (PC)

File: Local Monitor Dashboard-beta1.1.1.py (PySide6)

6.1 Tab aplikasi

Tab	Fungsi utama
Map Points	Klik peta → waypoint; Set Home; tabel marker; Send to Remote; panel Setup
Live Data	Peta tracking; indikator; 3 plot time-series; log CSV; route WP; koreksi rudder
Analyze Data	Load CSV; peta replay; plot; indikator; slider timeline

6.2 Map Points — panel kanan (urutan)

1. **Live position** — lat/lon/heading dari serial terbaru
2. **Set Home Point** — Home dari posisi live (butuh connect)
3. **Marker Points** — tabel 11 baris (Home + 10 WP max tampilan)
4. **Send to Remote** — info jumlah point, tombol kirim, status WP/TUN/verify
5. **Setup** — tombol **Open Setup** → dialog alg + tuning

6.3 Setup dialog

- **Alg 1** — waypoint + PD; spinbox Kp, Kd, arrive (m), rudder max (°)
- **Alg 2** — stub (default); tanpa parameter
- **Read from Remote** — \$TUNGET, update UI dari \$TACK

Nilai dialog disimpan lokal; commit ke kapal hanya via **Send to Remote**.

6.4 Send to Remote — state machine

Fase	Kirim	Respons sukses	Status UI
wp	\$WPSET, . . .	\$WACK, OK, WP	Lanjut TUN
tun	\$TUNSET, . . .	\$WACK, OK, TUN	Lanjut TUNGET
tunget	\$TUNGET	\$TACK, . . .	Verified OK / MISMATCH

Validasi sebelum kirim:

- Serial connected
- home_point_coords sudah di-set
- Minimal **Home + 2 waypoint** (total ≥ 3 point)
- Semua koordinat dalam range valid

Snapshot waypoint otomatis ke waypoints/DDMMYYYY_HHMM_WayPoints.csv.

6.5 Log CSV (tab Live)

Header TELEMETRY_LOG_HEADER — nilai **tampilan** (bukan raw fixed-point), sama seperti panel Live.

6.6 Analyze — format CSV didukung

Format	Deteksi
display_v23	Kolom speedMps (m/s), heading_setpoint (°)
raw_v23	Kolom heading_setpoint (raw)
legacy	Format lama

6.7 Dependencies Python

Python 3.10+, PySide6, pyserial, PyQtWebEngine.

```
cd Pythonfile/Way_Points_Tracking python "Local Monitor Dashboard-beta1.1.1.py"
```

7. Protokol serial (Dashboard ↔ User-Side)

Semua perintah diakhiri \n. Baud: **115200**.

7.1 \$WPSET — set waypoint

```
$WPSET,<home_lat>,<home_lon>,<wp_count>,<lat1>,<lon1>,...,<latN>,<lonN>
```

Field	Aturan
home_lat, home_lon	double, ±90° / ±180°
wp_count	0–10
waypoint	pasangan lat, lon per WP

Contoh (Home + 2 WP):

```
$WPSET,-7.281500,112.798900,2,-7.282000,112.799500,-7.283000,112.800000
```

7.2 \$TUNSET — set tuning

```
$TUNSET,<alg>[,<kp>,<kd>,<arrive_m>,<rudder_max>]
```

alg	Parameter	Contoh
1	4 float wajib	\$TUNSET,1,1.0000,0.0500,3.00,40.00
2	tidak ada	\$TUNSET,2

7.3 \$TUNGET — read-back tuning dari NVS Remote

\$TUNGET

7.4 Respons \$WACK

Pola	Makna
\$WACK, OK, WP	Waypoint diterima Remote
\$WACK, OK, TUN	Tuning disimpan NVS
\$WACK, ERR, <kind>, <reason>[, extra]	Gagal

7.5 Respons \$TACK

Pola	Makna
\$TACK, <alg>, <kp>, <kd>, <arrive>, <rudmax>	Read-back sukses
\$TACK, ERR, <reason>	Gagal

7.6 Tabel error User-Side (validasi lokal)

Waypoint (kind=WP):

Reason	Penyebab
FORMAT	Token tidak cukup
HOME_LAT, HOME_LON	Home tidak numerik
COUNT_NOT_INT	wp_count bukan integer
COUNT_RANGE	wp_count < 0 atau > 10
COUNT_MISMATCH	jumlah token $\neq 3 + 2 \times \text{wp_count}$
LAT_RANGE, home / LON_RANGE, home	Home out of range
WP_LAT, n / WP_LON, n	WP tidak numerik
LAT_RANGE, n / LON_RANGE, n	WP out of range
SEND_FAIL	esp_now_send gagal
TIMEOUT	Remote tidak ACK dalam 2.5 s
LINE_TOO_LONG	Baris serial terlalu panjang

Tuning (kind=TUN):

Reason	Penyebab
FORMAT	Token kosong

ALG	alg tidak integer
ALG_RANGE	alg bukan 1 atau 2
PARAM_COUNT	alg=1 tapi param ≠ 4
KP, KD, ARRIVE, RUDMAX	float tidak valid
SEND_FAIL, TIMEOUT	sama seperti WP

TACK:

Reason	Penyebab
SEND_FAIL	esp_now_send gagal
TIMEOUT	Remote tidak balas 0xA3

7.7 Error ACK dari Remote (via \$WACK, ERR, TUN, <code>)

err_code	Makna
1 (VALIDATE)	Parameter tuning out of range
2 (NVS)	Gagal simpan Preferences

8. Protokol ESP-NOW (User-Side ↔ Remote-Side)

msg_type	Arah	Struct	Ukuran
0xA1	User → Remote	waypoints_payload	~180 B
0xA2	User → Remote	track_config_payload	20 B
0xB1	User → Remote	tun_get_request	1 B
0xA3	Remote → User	track_config_payload	20 B
0xC1	Remote → User	remote_ack_payload	4 B
telemetry	Remote → User	DatatoSend / receivedfromremoteside	64 B

track_config_payload

```
msg_type, active_alg, param_count, reserved, params[0]=Kp, params[1]=Kd, params[2]=arrive_m,
params[3]=rudder_max_deg
```

remote_ack_payload

```
msg_type=0xC1, ack_kind (1=WP, 2=TUN), status (0=OK, 1=ERR), err_code
```


9. Telemetry 23 kolom

Urutan CSV dari User-Side (raw fixed-point):

```
timestamp, latitude, longitude, speedMps, Calc_deg_servo_1, Calc_deg_servo_2,  
yaw, heading_setpoint, heading_error, rudder_cmd, track_wp_index, distance_to_wp,  
accel_x, accel_y, accel_z, gyro_x, gyro_y, gyro_z, rpm_prop_1, rpm_prop_2, battery_1, battery_2, mode_auto
```

#	Field	Tipe Remote	Skala dashboard	Unit tampilan
1	timestamp	double	asli	s
2	latitude	double	asli	°
3	longitude	double	asli	°
4	speedMps	uint16	÷100	m/s
5	Calc_deg_servo_1	int16	÷100	°
6	Calc_deg_servo_2	int16	÷100	°
7	yaw	uint16	÷100	° (0–360)
8	heading_setpoint	uint16	÷100	°
9	heading_error	int16	÷100	°
10	rudder_cmd	int16	÷100	° offset
11	track_wp_index	uint8	asli	0=—, 1..N=WP, 255=Home
12	distance_to_wp	uint16	÷10	m
13–15	accel_x/y/z	int16	÷100	g
16–18	gyro_x/y/z	int16	÷100	deg/s
19–20	rpm_prop_1/2	uint16	÷100	rpm
21–22	battery_1/2	uint16	÷100	V
23	mode_auto	uint8	asli	0/1/2

Update rate: ~10 Hz (loop Remote 100 ms).

10. Algoritma auto track

10.1 Alg 1 — waypoint + PD

Persyaratan: GPS valid, waypoint sudah diterima (\$WPSET sukses), CH6 ≥ 1750 .

Urutan target:

1. Jika `wp_count > 0`: navigasi ke WP1 $\rightarrow$ WP2 $\rightarrow$... (index `g_active_wp_index`)
2. Jika `wp_count == 0` dan `home_valid`: target Home (`track_wp_index = 255`)
3. Saat jarak ke WP aktif $< arrive_m$, advance ke WP berikutnya

Kontrol:

```
bearing = bearingDeg(pos, target) heading_error = wrap(bearing - yaw,  $\pm 180^\circ$ ) rudder_offset = Kp  $\times$  heading_error - Kd  $\times$  gyro_z rudder_offset = clamp(rudder_offset,  $\pm rudder\_max$ )
```

Telemetry navigasi: `heading_setpoint = bearing, heading_error, track_wp_index, distance_to_wp`.

10.2 Alg 2 — stub (default)

Rudder netral; `heading_setpoint, heading_error, track_wp_index, distance_to_wp = 0`.

10.3 Parameter tuning Alg 1

Parameter	Default	Range validasi Remote
Kp	1.0	0 – 10
Kd	0.05	0 – 2
WP arrive (m)	3.0	0.5 – 50
Rudder max (°)	40.0	1 – 45

11. NVS tuning (Remote)

Item	Nilai
Namespace	<code>wptrack</code>
Magic	<code>0xA24150 (TUN_CFG_MAGIC)</code>
Keys	<code>magic, alg, kp, kd, arrive, rudmax</code>

Load saat `setup()` via `loadTrackConfigFromNvs()`. Jika magic tidak cocok $\rightarrow$ default Alg 2.

Waypoint **tidak** disimpan NVS (hanya RAM sampai power cycle atau \$WPSET baru).

12. Konfigurasi MAC ESP-NOW

PENTING: MAC harus saling cocok di kedua firmware.

Remote `src/main.cpp`:

```
uint8_t user_side_Address[] = {0x80, 0xb5, 0x4e, 0xc1, 0xd5, 0xac};
```

User-Side src/main.cpp:

```
uint8_t remote_side_Address[] = {0x10, 0x20, 0xba, 0x4c, 0x53, 0xfc};
```

Cek MAC perangkat: upload firmware yang print MAC di setup(), atau tool WiFi scanner. Sesuaikan pasangan di kedua file sebelum uji lapangan.

13. Build, upload & menjalankan

Remote-Side

```
cd PlatformIO/Way_Points_Tracking/ESP-Now_ESP32-S3_Remote-Side-01 pio run pio run --target upload  
pio device monitor
```

User-Side

```
cd PlatformIO/Way_Points_Tracking/ESP-Now_ESP32-S3_User-Side-01 pio run pio run --target upload pio  
device monitor
```

Sesuaikan upload_port / monitor_port di platformio.ini tiap proyek.

Dashboard

```
cd Pythonfile/Way_Points_Tracking python "Local Monitor Dashboard-beta1.1.1.py"
```

14. Prosedur uji lapangan

1. **Flash** Remote-Side ke ESP di kapal, User-Side ke ESP gateway USB
2. **Verifikasi MAC** ESP-NOW cocok di kedua main.cpp
3. **Power** kapal — tunggu GPS fix (monitor Remote optional)
4. **Connect** dashboard ke port User-Side (115200)
5. Tab **Map Points** → tunggu lat/lon live update
6. **Set Home Point** dari posisi aktual
7. Klik **2+ waypoint** di peta
8. Panel **Setup** → pilih alg & param (default Alg 2)
9. **Send to Remote** → status harus **Verified OK**
10. Opsional: **Read from Remote** di Setup untuk cek NVS
11. **RC CH6 ≥ 1750** → mode Auto di kapal
12. Tab **Live Data** → pantau track_wp_index, distance_to_wp, rudder_cmd

15. Troubleshooting

Gejala	Kemungkinan penyebab	Tindakan
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